3. A first order differential equation of the form $\frac{dx}{dy}+Px=Q$, where P and Q are functions of y only. Here no product of x and its derivative $\frac{dx}{dy}$ occur and the dependent variable x and its derivative with respect to independent variable y occurs only in the first degree.

In this case, the solution is given by $xe^{\int Pdy}=\int Qe^{\int Pdy}dy+C$.

Example 10.22

Solve $\frac{dy}{dx}+2y=e^{-x}$.

Solution

$$\text{Given that } \frac{dy}{dx}+2y = e^{-x} \qquad \ldots (1)$$

This is a linear differential equation.

Here $P=2$; $Q=e^{-x}$.

$$\int Pdx = \int 2dx = 2x.$$

$$\text{Thus, I.F.} = e^{\int Pdx}=e^{2x}.$$

$$\text{Hence the solution of (1) is } ye^{\int Pdx} = \int Qe^{\int Pdx}dx+C.$$

That is, $ye^{2x}=\int e^{-x}e^{2x}dx+C$ or $ye^{2x}=e^{x}+C$ or $y=e^{-x}+Ce^{-2x}$ is the required solution. ■

Example 10.23

Solve $\left[y\left(1-x\tan x\right)+x^2\cos x\right]dx-xdy=0$.

Solution

The given equation can be rewritten as $\frac{dy}{dx}+\frac{\left(x\tan x-1\right)}{x}y=x\cos x$.

This is a linear differential equation. Here $P=\frac{\left(x\tan x-1\right)}{x}$; $Q=x\cos x$.

$$\int Pdx=\int\frac{\left(x\tan x-1\right)}{x}dx=-\log|\cos x|-\log|x|=-\log|x\cos x|=\log\frac{1}{|x\cos x|}.$$

$$\text{Thus, I.F.} = e^{\int Pdx}=e^{\log\frac{1}{|x\cos x|}}=\frac{1}{x\cos x}$$

$$\text{Hence the solution is } ye^{\int Pdx} = \int Qe^{\int Pdx}dx+C$$

$$\text{i.e., } y\frac{1}{x\cos x} = \int\left(x\cos x\right)\frac{1}{x\cos x}dx+C$$

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